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Characterization of Reference Materials for *DPYD*: A GeT-RM Collaborative Project

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Abstract

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The *DPYD* gene encodes dihydropyrimidine dehydrogenase (DPD) which is involved in the catalysis of uracil and thymine as well as 5-fluorouracil (5-FU), which is used to treat solid tumors. Patients with decreased DPD activity are at risk of serious, sometimes fatal, adverse drug reactions to this important cancer drug. Pharmacogenetic testing for *DPYD* is increasingly provided by clinical and research laboratories; however, only a limited number of quality control and reference materials are currently available for clinical *DPYD* testing. To address this need, the Division of Laboratory Systems, Centers for Disease Control and Prevention (CDC) based Genetic Testing Reference Materials Coordination Program (GeT-RM), in collaboration with members of the pharmacogenetic testing and research communities and the Coriell Institute for Medical Research, has characterized 33 DNA samples derived from Coriell cell lines for *DPYD*. Samples were distributed to four volunteer laboratories for genetic testing using a variety of commercially available and laboratory-developed tests. Sanger sequencing was utilized by one laboratory and publicly available whole genome sequence (WGS) data from the 1000 Genomes Project was utilized by another to inform genotype. Thirty-three distinct *DPYD* variants were identified among the 33 samples characterized. These publicly available and well-characterized materials can be used to support the quality assurance and quality control programs of clinical laboratories performing clinical pharmacogenetic testing.

Introduction

The *DPYD* gene, located on chromosome 1p22, is a large gene with 23 exons spanning 843 kb (NCBI <https://www.ncbi.nlm.nih.gov/gene?Cmd=DetailsSearch&Term=1806>; last accessed February 15, 2024). *DPYD* encodes dihydropyrimidine dehydrogenase (DPD) which is involved in the catabolism of the pyrimidine bases uracil and thymidine¹ as well as 5-fluorouracil (5-FU), which is commonly used to treat solid tumors.² Individuals with variants in *DPYD* that decrease or eliminate DPD activity have an increased risk of toxicity from 5-FU, as well as from 5-FU prodrugs capecitabine and tegafur, which can cause severe or fatal adverse drug reactions (ADRs), including bone marrow suppression, gastrointestinal toxicity, and neurotoxicity.^{3–5}

Approximately 1600 *DPYD* variants are currently listed in gnomAD⁶, a subset of which are included in the Pharmacogene Variation Consortium (PharmVar)⁷ (<https://www.pharmvar.org/gene/DPYD>; last accessed 1/26/2024). While the function of many variants remains unknown or uncertain, others have been shown to account for significant inter-individual variation in enzyme activity, which can profoundly affect how patients respond to drugs metabolized by DPD. Dosing guidelines based on *DPYD* genotype have been published by the Clinical Pharmacogenetics Implementation Consortium (CPIC)⁸ (<https://cpicpgx.org/guidelines/guideline-for-fluoropyrimidines-and-dpyd/>; last accessed 12/7/2023), the Dutch Pharmacogenetics Working Group (DPWG)⁴, the Spanish Society of Pharmacogenetics and Pharmacogenomics (SEFF)⁹, and the French National Network of Pharmacogenetics (RNPGx)¹⁰.

Clinical laboratories offer tests that can detect specific variants in pharmacogenes, including *DPYD*, which can then be used to predict or explain an individual's metabolizer status. Prescribers can use the results of pharmacogenetic tests to select an appropriate drug

and/or dose for each patient to ensure effective treatment and avoid or minimize ADRs. However, clinical *DPYD* tests may differ substantially in the *DPYD* variants tested and their analytical-methodological approaches. Furthermore, there are few publicly available, well-characterized reference materials to support test development and validation studies.

To address the lack of standardization in pharmacogenetic test panels, the Association for Molecular Pathology (AMP) Pharmacogenetic Working Group has developed a series of documents that recommend a minimum set of variant alleles to include in clinical pharmacogenetic test panels^{11–16} including *CYP3A4/3A5*, *CYP2D6*, *CYP2C19*, *CYP2C9* and others. Most recently, the Working Group has developed recommendations for clinical *DPYD* testing.¹⁷ The AMP Pharmacogenetic Working Group has established four criteria that must be met for alleles to be recommended for inclusion in clinical testing panels. One of these criteria is the availability of reference materials.

To support development and validation of quality *DPYD* genetic tests, the Division of Laboratory Systems, Centers for Disease Control and Prevention (CDC) based Genetic Testing Reference Materials Coordination Program (GeT-RM), the Coriell Institute for Medical Research, and the genetic testing community have collaborated to characterize genomic DNA samples from 33 publicly available cell lines for use as *DPYD* reference materials in clinical testing. The goal of this GeT-RM study was to create characterized genomic DNA reference materials for as many *DPYD* variants as possible.

Materials and Methods

Participating Laboratories

Four laboratories, utilizing a variety of methods and test platforms, participated in this effort: Children's Mercy Research Institute (CMRI, Kansas City, MO; Laboratory 1), RPRD Diagnostics (Milwaukee, WI; Laboratory 2), Mayo Clinic (Rochester, MN; Laboratory 3), and the Hospital Universitario de la Princesa, Madrid (Madrid, Spain; Laboratory 4).

Cell lines DNA samples

DNA from 32 cell lines were selected from the National Institute of General Medical Sciences (NIGMS) Human Genetic Cell Repository and the National Human Genome Research Institute (NHGRI) Sample Repository for Human Genetic Research at the Coriell Institute for Medical Research (Camden, NJ) based on data supplied by the authors or identified by searching the high-coverage WGS data from the 1000 Genomes Project (1kGP).¹⁸ Samples with *DPYD* variants of interest were identified from the 1kGP using the Ensembl browser (<https://useast.ensembl.org/index.html>; last accessed 12/7/2023) and by Laboratory 1 using a combination of BCFtools version 1.11 and Variant Effect Predictor (VEP) version 109.^{19–21} The variant list was filtered for samples having nonsynonymous single nucleotide variants (SNVs) within coding regions, SNVs within exon/intron junctions (splice recognition sites), and/or a deep intronic SNV, NM_000110.4:c.1129–5923C>G, that is part of the HapB3 haplotype. In addition, the Progenetix search tool (<https://progenetix.org/progenetix-cohorts/oneKgenomes/>; last accessed 1/3/2024) was utilized to identify 1kGP samples with *DPYD* copy number variation (CNVs). An additional sample

was added during this investigation for a total of 33 samples. Laboratory 2 had purchased DNA and was able to test a variant of interest that had a low-quality WGS call and was also not identified during the initial Ensembl screen.

If there were several samples harboring an SNV(s) of interest, two were chosen for further follow-up with a preference for samples that have been included in a previous GeT-RM study.²² Selected samples were subsequently tested by the laboratories to confirm presence of SNV(s) (Laboratories 2, 3, and 4) and exonic deletions (laboratories 1–3). The sample added during this investigation was analyzed by laboratory 2, which had the DNA sample available for testing.

DNA was prepared from each of the selected cell lines by the Coriell Institute for Medical Research using Gentra/Qiagen Autopure (Valencia, CA) per manufacturer's instructions.

Characterization Protocol

Laboratories 1–4 received one 10 µg aliquot of DNA from 32 cell lines and tested all or a subset of the samples using their standard methods of variant and/or CN (copy number) detection. Laboratories 1–3 experimentally interrogated deletion events involving two regions. The testing platforms and assays used in the study are described below and the genotyping assays are summarized in Table 1. Results were submitted to authors LVK and AG for examination of the data for quality, discordances, and determination of consensus genotype.

Allele Designations

Variant positions are provided throughout this report as recommended by the Human Genome Variation Society (<https://www.hgvs.org/> last accessed 2/15/2024) using NM_000110.4 and NP_000101.2 as reference sequences (RefSeq <https://www.ncbi.nlm.nih.gov/refseq/>, last accessed 5/17/2024); rsIDs (dbSNP <https://www.ncbi.nlm.nih.gov/snp>, last accessed 2/2/2024) are provided for each variant when first mentioned in the text and are provided in all tables including supplemental materials. PharmVar lists variants using their rsIDs as “Allele Names” (<https://www.pharmvar.org/gene/DYPD>; last accessed 02/15/2024). Of importance, previously used *DPYD* star allele designations do not describe haplotypes but rather individual SNVs. Although star allele-based nomenclature was initially proposed by McLeod and colleagues²³, the PharmVar expert panel recommended abandoning phased haplotype (star allele) definitions for *DPYD* in favor of listing individual variants. Single nucleotide polymorphisms (SNPs) and small insertions and deletions (indels) are collectively referred to herein as single nucleotide variants (SNVs).

Laboratory 1 (CMRI)

1000 Genomes Project (1kGP) whole genome sequence (WGS) data

analysis: WGS data of the 1kGP were utilized to create a variant list as described above.

Copy number variation testing for exon 4 and exon 11—Six Coriell DNAs were tested for the presence of *DYPD* exon 4 and exon 11 deletions on two platforms, the

QX200 Droplet Digital PCR System (Bio-Rad, Hercules, CA, USA) and the Absolute Q system (Thermo Fisher Scientific, Waltham, MA). Each platform utilized pre-designed TaqMan copy number assays (Thermo Fisher Scientific, Waltham, MA, USA) (Table 1). Assays were run in both single target reactions and duplexed by amplitude. DNA was initially digested with Anza 69 BglI (Invitrogen, Waltham, MA, USA) using 50 ng of DNA according to manufacturer's protocol and applied to both single-target and duplexed formats. Single-target reactions (total volume 21 μ L) were mixed with nuclease-free water for a final concentration of 1x ddPCR Supermix for Probes (no dUTP), 1.0x *DPYD* TaqMan copy number assay, 1.0x RNaseP TaqMan copy number reference assay (Cat# 4403326), and 12.5 ng of digested DNA. Amplitude-multiplexed reactions were combined similarly, except for the final concentrations of the following TaqMan Assays: 1.0x *DPYD* exon 4, 0.5x *DPYD* exon 11, and 1.0x RNaseP reference. Subsequently, droplets were generated using an AutoDG Instrument, sealed with pierceable foil, and thermocycled according to the manufacturer's protocol using a C1000 touch (Bio-Rad, Hercules, CA, USA). Results were manually analyzed and inspected using QuantaSoft Analysis Pro (v1.0.596).

On the Absolute Q dPCR System (Applied Biosystems, Waltham, MA, USA) copy number was determined using single-target reactions. DNA was digested using the previously established One-pot method (Wang *et al.* 2024; manuscript submitted), where all reagents are combined in a single step for digestion and CNV determination. Reaction mixes of 10 μ L were combined with nuclease-free water for final concentrations of 1x Absolute Q DNA Digital PCR Master Mix, 1x *DPYD* TaqMan copy number assay, 1x RNaseP TaqMan reference assay with custom ABY fluorescent label, 1x Anza Clear Buffer, 1x Anza 69 BglI enzyme, and 10 ng of DNA. Reaction mixes were then loaded onto an Absolute Q MAP16 Plate, incubated at ambient temperature for 30 minutes, and subsequently run on the Absolute Q instrument according to the manufacturer's protocol. Results were manually analyzed and inspected using the Absolute Q Digital PCR System Software (version 6.2.1).

Laboratory 2 (RPRD)

Genotyping was performed as previously described using the PharmacoScan™ Assay Kit, catalog ID 903010 (Thermo Fisher Scientific, Waltham, MA) following manufacturer's instructions.²⁴ Arrays were hybridized, stained with fluorescent antibodies, and scanned on the GeneTitan™ Multi-Channel (MC) Instrument (Thermo Fisher Scientific, Waltham, MA). Data were analyzed using the Axiom™ Analysis Suite 5.1.1.1 (Thermo Fisher Scientific, Waltham, MA). Genotype calls were made using the commercially released allele translation table (r9). Variants tested by the PharmacoScan platform are summarized in Table 1.

Laboratory 3 (Mayo Clinic)

DNA samples were analyzed for variants in *DPYD* by Sanger sequencing, copy number testing by quantitative real-time PCR (qPCR), and/or targeted genotyping, all using GRCh37 NM_000110.3 as the reference sequence. Sanger sequencing was performed for all exons and intron/exon junctions (+/-10 bp) as well as the c.1129-5923C>G variant. BigDye Terminator chemistry version 1.1 and an Applied Biosystems International (ABI) PRISM 3730xl DNA Analyzer (Thermo Fisher Scientific, Waltham, MA) were used for sequencing,

followed by analysis with Mutation Surveyor version 4.0.9 (Soft Genetics, State College, PA).

A ViiA 7 instrument was used for qPCR (Thermo Fisher Scientific). Custom primers were designed for exons 4 and 11, along with adjacent exons to serve as a control (exons 3, 5, 10, and 12). The region of interest in *DPYD* was amplified along with the control genes *GAPDH* and *CLCN7*, followed by detection of double stranded PCR products by SYBR Green intercalation. Crossing threshold (CT) values for the *DPYD* amplicons were compared to the control amplicons, and samples known to have two copies of *DPYD*.

Genotyping was performed using TaqMan chemistry in a custom-designed Open Array format on a QuantStudio 12K Flex instrument (Thermo Fisher Scientific) with assays designed to detect eight *DPYD* variants as indicated in Table 1. Analysis was performed using Genotyper software version 1.2.2 (Thermo Fisher Scientific) and a custom-designed proprietary software, GINger version 1.0 (Mayo Clinic, Rochester, MN).

Laboratory 4 (Hospital Universitario de la Princesa)

Genotyping was performed as previously described²⁵ using the Very Important Pharmacogene Open Array (10.3390/jcm10173772) version 4, a customized Open Array plate run on a QuantStudio 12k Flex qPCR instrument (Thermo Fisher Scientific, Waltham, MA) that contains nine *DPYD* TaqMan assays as indicated in Table 1 (Thermo Fisher Scientific, Waltham, MA); seven of the assays are commercially available and two are custom designed.

Results

DNA from the 32 selected cell lines was tested by four laboratories for *DPYD* using a variety of genotyping and sequencing methods. An additional DNA sample was added later in the study and WGS data was experimentally confirmed by Laboratory 2. CNVs were interrogated by three laboratories. Testing confirmed the SNVs identified using the 1kGP WGS data if the testing platforms covered/interrogated the variant. Results from each laboratory are summarized in Supplemental Table 1.

A total of 33 *DPYD* variants were identified in the selected samples. These include 26 nonsynonymous SNVs, two SNVs which cause aberrant splicing and two exon deletion events. Two samples have the deep intron 10 variant c.1129–5923C>G rs75017182 and the synonymous exon 11 variant c.1236G>A, p.Glu412=, rs56038477 (formerly HapB3). One sample, HG03787, has c.2043_2058del, p.Leu682IlefsTer24, rs773499329 which was not among those listed by PharmVar or CPIC at the outset of this study; this variant was submitted to PharmVar and is now posted. The presence of this variant was confirmed by Laboratory 3 using Sanger sequencing. The variant has almost exclusively been found in South Asians at a frequency of 0.099% (https://gnomad.broadinstitute.org/variant/1-97373560-CCTGCCCCACAGGCCAGG-C?dataset=gnomad_r4; last accessed 02/15/2024). In addition, there were two samples with exonic deletion events. Specifically, HG00325 had a deletion comprising exon 4 and adjacent intronic sequences and NA19093 had a deletion encompassing exon 11 and adjacent intronic sequences. The presence of these

deletions was confirmed using quantitative copy number testing, namely digital PCR and quantitative real-time PCR. As shown in Supplemental Figure 1, the exon 11 deletion was also detected using PharmacoScan SNV signal ratios; however, this platform does not allow detection of the exon 4 deletion as the array does not interrogate any SNVs in this gene region. Additionally, as shown in Supplemental Figure 2, Laboratory 1 detected a secondary cluster on digital PCR scatter plots when interrogating the exon 11 deletion of HG00613 in both the single-target and duplexed reactions. This sample has a rare heterozygous variant in exon 11 (c.1314T>G, p.Phe438Leu, rs186169810) which interferes with the assay (personal communication with Thermo Fisher Scientific application specialist) and causes the observed secondary cluster. However, the presence of the additional cluster did not affect copy number call (CN=2) when gated correctly. Of note, this variant was found in *cis* with c.1627A>G (*5) based on trio analysis.

Thirteen of the 33 samples also harbored one or more synonymous SNVs, c.1371C>T, p.Asn457=, rs57918000 and c.1896T>C, p.Phe632=, rs17376848, or c.1236G>A, p.Glu412=, which were found in two, nine, and three samples, respectively. Although these SNVs were not among those for which reference materials were sought, they are tabulated in Supplemental Tables 1 and 2 for completeness.

Three samples (NA06991, NA18966, and NA19207) were characterized for *DPYD* in a previous GeT-RM study.²² NA06991 was found to be heterozygous for c.1601G>A, p.Ser534Asn, rs142619737 (*4), confirming the presence of this SNV which was suspected, but not previously confirmed. This sample was also found to be heterozygous for c.2846A>T, p.Asp949Val, rs67376798. Sample NA18966 was previously reported as *DPYD**1/*1²² but was found to be heterozygous for both the c.1896T>C, p.Phe632=, rs17376848 and c.2678A>G, p.Asn893Ser, rs188052243 variants. Finally, sample NA19207 was reported as *DPYD**1/*9²² but was found to not only be heterozygous for c.85T>C, p.Cys29Arg, rs1801265 but also for the c.557A>G, p.Tyr186Cys, rs115232898 variant.

Sample HG01631 was not identified by the initial Ensembl search of the 1kbGP dataset as having c.299_302del, p.Phe100SerfsTer15, rs72549309 (*7). This variant is called by WGS but has a Variant Quality Score Recalibration (VQSR) value raising concerns regarding the validity of the call. Testing by laboratory 2, however, confirmed the presence of the c.299_302del (*7) nucleotide deletion in sample HG01631. This example underscores the importance of validating variants of interest with orthogonal methods to unequivocally demonstrate presence or absence of variants.

Consensus variant calls

Consensus calls are provided on a per variant basis and not for the entire complement of variants found in a sample (Table 2 and Supplemental Table 1). For 31 of the 33 study samples there was consensus across all variants because all SNVs present in the WGS data set were experimentally confirmed by at least one method.

Two samples had one WGS-identified SNV that was not experimentally confirmed because the assays used to test the sample did not interrogate the variant; these are shown in brackets in Supplemental Table 1 and indicated in Table 2. The non-confirmed SNVs do not represent

those for which these samples were selected and are not considered clinically significant. Specifically, HG03115 was chosen because WGS indicated that this sample has c.61C>T, p.Arg21Ter, rs72549310, which was experimentally confirmed. However, HG03115 is also heterozygous for the synonymous variant c.1371C>T, p.Asn457=. Since this SNV is not interrogated by the PharmacoScan assay and the sample was not selected for Sanger sequencing, there are no experimental data. NA19093 was chosen to confirm the presence of an exon 11 deletion event. This sample, per WGS, also has c.85T>C, p.Cys29Arg; c.1218G>A, p.Met406Ile, rs61622928 and c.1627A>G, p.Ile543Val, rs1801159. Since the PharmacoScan does not interrogate c.1218G>A, and the sample was not selected for Sanger sequencing, this variant was not experimentally confirmed either. Consensus was, however, obtained for the exon 11 deletion and the other two SNVs.

***DPYD* haplotypes**

Of the 33 samples tested in this study, six (HG00325, HG00332, HG03770, NA18956, NA20786 and NA20901) had a single variant, one (NA19711) was homozygous for all but one variant, and two samples (HG01374 and HG03649) were homozygous for all SNVs allowing haplotypes to be inferred. Because *DPYD* is a large gene spanning over 800 kb, it was not possible to identify haplotypes using Sanger or short read WGS data. Trio information (inheritance) was available to conclusively infer haplotypes for 12 samples (HG00537, HG00538, HG00613, HG01350, HG01631, HG02645, HG02772, HG03115, NA06991, NA12248, NA19093, NA19207) (Supplemental Table 2). Haplotypes could not, or only partially, be determined for 12 samples (HG00118, HG00129, HG00185, HG00315, HG01253, HG03645, HG03716, HG03787, NA18966, NA19900, NA19902, and NA20362) (Supplemental Table 2). c.1129–5923C>G and c.1236G>A are assumed to be in *cis* as c.1129–5923C>G has never been described to occur without c.1236G>A. In contrast c.1236G>A has, in rare cases, been found without c.1129–5923C>G.^{26, 27} Two normal function variants, namely c.85T>C, p.Cys29Arg (*9A) and c.1627A>G, p.Ile543Val (*5) are both found in *cis* (i.e., on the same haplotype) with SNVs that have been associated with altered function or unknown function; hence, these variants, although having normal function, may reveal associations with decreased DPD function or toxicity.

Discussion

Characterized reference materials are used for various quality assurance purposes, including test development, validation, quality control and proficiency testing/alternative assessment. The use of reference materials is also mandated by regulations, accreditation standards, and professional guidance^{28–32} (American College of Medical Genetics and Genomics <https://www.acmg.net/PDFLibrary/ACMG%20Technical%20Lab%20Standards%20Section%20G.pdf>; last accessed 12/21/2023, Washington State Legislature, <http://app.leg.wa.gov/WAC/default.aspx?cite=246-338-090>; last accessed 12/21/2023, College of American Pathologists (Northfield, IL), New York State Clinical Laboratory Evaluation Program, <https://www.wadsworth.org/regulatory/clep>; last accessed 12/21/2023, MMWR <https://www.cdc.gov/mmwr/preview/mmwrhtml/rr5806a1.htm>; last accessed 12/21/2023). Despite these requirements, there are few, if any, reference materials available for most clinical genetic tests, including *DPYD*.

To address this need, the Centers for Disease Control and Prevention's Genetic Testing Reference Material Program (GeT-RM <https://www.cdc.gov/labquality/get-rm/index.html>; last accessed 12/7/2023) has conducted a number of projects to create characterized and publicly available cell line-based DNA samples for use as reference materials, including for pharmacogenetic testing.^{22, 24, 33–37}

Accurate *DPYD* genetic testing prior to treatment with 5-FU is critical for patient care. As more is learned about this very polymorphic gene, attention is focused on providing guidance, such as those from the AMP PGx Work Group¹⁷ to laboratories about which alleles to include in targeted genotype assays. Provision of characterized genomic DNA reference materials for as many important *DPYD* alleles as possible is needed to facilitate development, validation, and quality assurance of clinical *DPYD* tests and is critical for implementing these guidelines. In this study, we characterized 33 publicly available DNA samples and identified a total of 33 *DPYD* variants including the most common decreased or no function variants, and two alleles with exonic deletions. The reference materials developed as part of this study will not only provide important resources for quality control, proficiency testing, and research, but also support the development and validation of new pharmacogenetic tests that conform to professional recommendations and regulatory requirements.

One variant identified as part of this study, c.2043_2058del, p.Leu682IlefsTer24 encodes a 16-bp TGCCCA(CAGGC)₂ deletion which likely inactivates the protein. This variant, with a frequency of 0.099% in South Asians (https://gnomad.broadinstitute.org/variant/1-97373560-CCTGCCCCACAGGCCAGG-C?dataset=gnomad_r4; last accessed 02/15/2024), was not listed by PharmVar or CPIC at the outset of this study. Although this variant meets criteria for inclusion in Tier 2, the AMP PGx Working Group ultimately did not recommend testing of this variant because it was not included in the list of variants previously curated by CPIC.¹⁷

Nomenclature and Haplotype

Because the utilization of star (*) nomenclature based on haplotypes was deemed impractical by the PharmVar gene expert panel for this gene due to its size (843 kb), PharmVar is using rsIDs and HGVS nomenclature instead of star names to facilitate standardized reporting. rsIDs indicate the position of a genetic change but do not specify the nucleotide change and thus may not be specific enough on their own for routine use without the inclusion of additional information. In the past, several *DPYD* SNVs were published with a star name. In contrast to the haplotype-based star allele designation concept, *DPYD* star names do not represent a haplotype but rather refer to a specific SNV. These “legacy” star names (*DPYD**2 through *13) are cross-referenced in the legacy label column on the PharmVar *DPYD* page (<https://www.pharmvar.org/gene/DPYD>; last accessed 1/4/2024). In addition to star names, the combination of c.1129–5923C>G and c.1236G>A, p.Glu412= is commonly known as the “HapB3” haplotype. While the synonymous variant c.1236G>A, p.Glu412= is believed to be benign, c.1129–5923C>G is a deep intronic variant which introduces a cryptic splice site causing aberrant splicing and decreased function.^{38–40} Recently, however, Turner *et al*²⁶ reported that these two variants are not 100% linked across

all populations as previously assumed, and Dierks *et al.*²⁷ describe a clinical case who was classified as intermediate metabolizer based on c.1236G>A testing only. The patient did not have the c.1129–5923C>G variant explaining well below target 5-FU levels.

There is a knowledge gap regarding the function of *DPYD* haplotypes, as they have not been systematically analyzed for the reasons explained above. The analytical-methodological difficulty of doing so at present could be resolved in the coming years with the advent of cheaper and more accessible long-read sequencing technologies that would allow variant phasing in such a large gene. This would make it possible to adapt the allelic nomenclature of *DPYD* to the star allele system, common to cytochrome P450 enzymes, *SLCO1B1*, *NUDT15*, etc. Subsequently, the function of the different haplotypes could be systematically addressed, and an update of reference materials would be required. However, until haplotypes can be accurately and reliably characterized and tested for, the use of star names is discouraged for this gene.

Except for HapB3, haplotype information is typically not considered for phenotype prediction. However, knowledge about haplotype structure may be informative when interrogating genetic associations with activity, toxicity, or other outcomes. To raise awareness of the fact that many variants do not occur in isolation, diplotypes were reconstructed as shown in Supplemental Table 2. Haplotypes could be fully or partially established for some of the samples in cases where haplotypes were either informative on their own (e.g., sample was homozygous for all variants or heterozygous for only one variant) or could be informed by inheritance using trio data. Although this analysis is limited, it clearly shows that several variants do occur on different haplotypes which may have implications not only on protein function but also association analyses.

Furthermore, a recent report describes a common germline variant (rs4294451) located 9,236 bp upstream of the ATG translation start codon, that alters the level of interactions between an enhancer and the *DPYD* promotor.⁴¹ Since this variant is outside the genomic and transcript RefSeqs, we refer to the variant as GRCh38:g.97930158T>A. Zhang *et al.* found that the reference “T” nucleotide (GRCh38:g.97930158T), which represents the less common allele (Global Allele Frequency T=0.27; <https://www.ncbi.nlm.nih.gov/snp/>; last accessed 4/12/2024) was associated with higher expression levels compared to the more common “A” allele (GRCh38:g.97930158A). Therefore, variants in coding regions and those that cause aberrant splicing may not be the only variation important for predicting DPD activity. While Zhang *et al.* describe the impact of rs4294451 on expression levels there is no information in their report regarding the linkage of rs4294451 with other variants within the gene, especially those commonly targeted by PGx testing. For example, alleles in the 1kGP data set with c.1236G>A and c.1129–5923C>G (HapB3) can have the GRCh38:g.97930158 “T” or “A” allele, raising the possibility that those with “T” may have overall higher expression levels and thus, more functional protein compared to those with the “A” allele. This could also explain why patients with c.1236G>A and c.1129–5923C>G (HapB3) experience variable levels of toxicity. To provide some limited information about this variant, (i.e., which of the interrogated samples have rs4294451T>A) and whether it is in *cis* or *trans* with other variants, this novel enhancer variant was added to Supplemental Table 2. Although these preliminary data are sparse, they clearly show that

the higher expression GRCh38:g.97930158 “T” allele is not restricted to haplotypes without any variants but can occur in *cis* with several variants, suggesting that haplotype structure for GRCh38:g.97930158T>A may add an additional layer of complexity when assessing its clinical relevance.

Variant Reporting

To ensure standardized *DPYD* reporting, Human Genome Variation Society (HGVS) nomenclature (<https://hgvs-nomenclature.org/stable/>; last accessed 3/6/2024) should be employed moving forward, and the use of legacy star designations or alternate description such as HapB3 should be phased out as it may not be clear whether this legacy name means that both variants, c.1236G>A and c.1129–5923C>G, were tested and are indeed both present. In this report, variants are denoted throughout according to HGVS using the NM_000110.4 transcript sequence as reference. Legacy names are provided, if applicable, for the sole purpose to facilitate cross-referencing.

Problems can occur when legacy *DPYD* star names alone are used to describe variants on clinical reports. As shown in the example in Figure 1, a patient heterozygous for the following SNVs: c.85T>C, p.Cys29Arg (*9A); c.496A>G, p.Met166Val; c.1129–5923C>G, (splice defect, HapB3); c.1236G>A, p. Glu412= (HapB3); and c.2194G>A, p. Val732Ile (*6) may be described as *6+*9A+HapB3, p.Met166Val essentially listing the star names of SNVs, if existing, the HapB3 name, and using the amino acid change (or transcript position) if no star name exists. Reporting the patient’s genotype like this may imply that the “*6+*9A+HapB3” variants are in *cis* and p.Met166Val is in *trans*. Also, although a “+” has been historically used in this context, use of a “+” may lead some end-users to believe that there has been a gene duplication or multiplication. Furthermore, intermixing legacy star names together with either cDNA (c.) or protein (p.) annotations can also be a source of confusion on *DPYD* test reports. Therefore, the aforementioned reporting (“*6+*9A+Hap3B, p.Met166Val” or similar) is not recommended. In many cases, laboratories may choose to only report functionally significant *DPYD* variants in which case some of the variants found in the example may be excluded from the test report. As the field works toward standard genotype reporting, inclusion of the predicted phenotype and activity score, which are typically reported with more consistency, may facilitate understanding and appropriate use of *DPYD* test results.^{22, 24, 28–37}

In conclusion, the reference materials described in this report (Table 2) will facilitate accurate clinical *DPYD* testing and serve as materials for quality control processes. Together, these characterized genomic DNA samples form a comprehensive set of reference materials for *DPYD* testing. GeT-RM will continue to work to establish cell lines and characterize additional variants in *DPYD*, and other PGx genes that lack reference materials. All reference materials developed by GeT-RM are publicly available from the NIGMS and NHGRI repositories at the Coriell Institute for Medical Research (Camden, NJ). More information on this and other reference material characterization projects is available at the GeT-RM website: <https://www.cdc.gov/labquality/get-rm/index.html>; last accessed 12/7/2023).

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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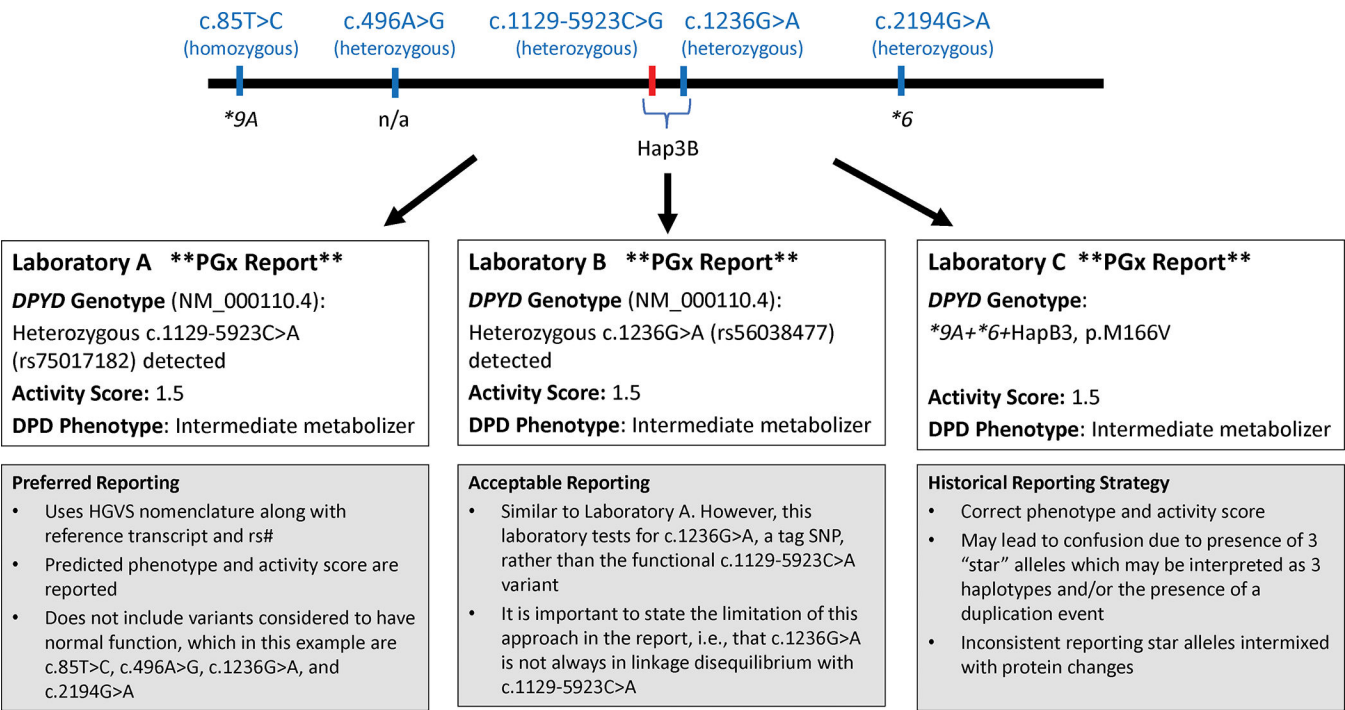


Figure 1.
Standardized reporting of *DPYD* variation
The sample shown has five SNVs, c.85T>C, p. Cys29Arg, rs1801265; c.496A>G, p. Met166Val, rs2297595; c.2194G>A, p. Val732Ile, rs1801160, and both variants of the HapB3 haplotype, c.1129–5923C>G (splice defect, rs75017182, highlighted by red line in top panel) and c.1236G>A, p. Glu412=, rs56038477. All variants are heterozygous, and their phase is unknown. Variants not causing any changes in DPD activity or variants of unknown function likely will not be included in panel testing; however, they may be discovered in sequencing-based approaches and may or may not be reported. While all three hypothetical laboratories (A, B and C) report the same activity score (1.5) and DPD phenotype (intermediate metabolizer), they differ in which variants are tested and how the *DPYD* genotype is reported.

Table 1.

Summary of Platforms and Genotyping Assays Used

Variant position NM_000110.4^ (Legacy Name)	rsID#	Laboratory 1 TaqMan assay ID%	Laboratory 2 PharmacoScan\$	Laboratory 3 TaqMan assay ID%	Laboratory 4 TaqMan assay ID%
c.40–3123T>G	rs4970722	NA	yes	NA	NA
c.61C>T	rs72549310	NA	yes	NA	NA
c.62G>A	rs80081766	NA	yes	NA	NA
c.85T>C (*9A)	rs1801265	NA	yes	NA	NA
c.151–69G>A	rs115632870	NA	yes	NA	NA
c.234–3075A>G	rs17378539	NA	yes	NA	NA
c.299_302del (*7)	rs72549309	NA	yes	AHHS7XT (custom)	AN9HUUX (custom)
c.496A>G	rs2297595	NA	yes	NA	NA
c.525G>T	rs6670886	NA	yes	NA	NA
c.557A>G	rs115232898	NA	yes	C_165900856_10	C_165900856_10
c.680+2545T>G	rs12119882	NA	yes	NA	NA
c.703C>T (*8)	rs1801266	NA	yes	C__8393861_20	C__8393861_20
c.756T>C	rs6675198	NA	yes	NA	NA
c.775A>G	rs45589337	NA	yes	NA	NA
c.850+26331C>T	rs2786507	NA	yes	NA	NA
c.851–18271A>G	rs2811196	NA	yes	NA	NA
c.851–38157C>T	rs10518636	NA	yes	NA	NA
c.1003G>T	rs72549306	NA	yes	NA	NA
c.1003G>A	rs72549306	NA	yes	NA	NA
c.1035T>C	rs1042478	NA	yes	NA	NA
c.1074T>A	rs1042479	NA	yes	NA	NA
c.1129–5923C>G (HapB3)	rs75017182	NA	yes	AHQJU5N (custom)	NA
c.1156G>T	rs78060119	NA	yes	NA	ANNK9ZJ (custom)
c.1236G>A (HapB3)	rs56038477	NA	yes	NA	C__25596099_30
c.1340–11501T>C	rs2811219	NA	yes	NA	NA
c.1601G>A (*4)	rs1801158	NA	yes	NA	NA
c.1627A>G (*5)	rs1801159	NA	yes	NA	NA
c.1679T>G (*13)	rs55886062	NA	yes	C__11985548_10	C__11985548_10
c.1740+8030G>T	rs17116806	NA	yes	NA	NA
c.1741–16477A>G	rs12136186	NA	yes	NA	NA
c.1741–2527G>A	rs3897854	NA	yes	NA	NA
c.1896T>C	rs17376848	NA	yes	NA	NA
c.1898del	rs72549303	NA	yes	NA	NA
c.1905C>T	rs3918289	NA	yes	NA	NA
c.1905C>G	rs3918289	NA	yes	NA	NA

Variant position NM_000110.4 [^] (Legacy Name)	rsID [#]	Laboratory 1 TaqMan assay ID [%]	Laboratory 2 PharmacoScan ^{\$}	Laboratory 3 TaqMan assay ID [%]	Laboratory 4 TaqMan assay ID [%]
c.1905+1G>A (*2A)	rs3918290	NA	yes	C__30633851_20	C__30633851_20
c.1906-14763G>A	rs12022243	NA	yes	NA	NA
c.1906-19696G>T	rs7548189	NA	yes	NA	NA
c.1906-28506C>G	rs4492658	NA	yes	NA	NA
c.1906-5426A>G	rs2152878	NA	yes	NA	NA
c.1974+75A>T	rs72728438	NA	yes	NA	NA
c.2194G>A (*6)	rs1801160	NA	yes	NA	NA
c.2300-23051C>T	rs2027056	NA	yes	NA	NA
c.2300-23459G>T	rs12140120	NA	yes	NA	NA
c.2622+9416C>T	rs6656660	NA	yes	NA	NA
c.2623-38806T>C	rs1760217	NA	yes	NA	NA
c.2623-42576G>C	rs7552825	NA	yes	NA	NA
c.2656C>T	rs147545709	NA	yes	NA	NA
c.2657G>A (*9B)	rs1801267	NA	yes	NA	NA
c.2767-2165A>T	rs290852	NA	yes	NA	NA
c.2767-5102A>G	rs17471125	NA	yes	NA	NA
c.2846A>T	rs67376798	NA	yes	C__27530948_10	C__27530948_10
c.2983G>T (*10)	rs1801268	NA	yes	C__8393505_20	C__8393505_20
NC_000001.11: g.97057448G>A	rs12132152	NA	yes	NA	NA
NC_000001.11: g.97073844G>A	rs76387818	NA	yes	NA	NA
Exon 4 Copy Number	NA	Hs03083443_cn	NA	NA	NA
Exon 11 Copy Number	NA	Hs03056809_cn	NA	NA	NA

[^]RefSeq <https://www.ncbi.nlm.nih.gov/refseq/>, last accessed 5/17/2024

[#]dbSNP (<https://www.ncbi.nlm.nih.gov/snp>, last accessed 2/2/2024)

[%]TaqMan (Thermo Fisher Scientific, Waltham, MA)

^{\$}PharmacoScan (Thermo Fisher Scientific, Waltham, MA)

NA, assay not performed

Yes, allele was tested

Table 2.Consensus *DPYD* genotypes

Coriell ID	Consensus Genotype (NM_000110.4) [^]	rsID ^{\$} (Legacy Name)
HG00118	c.85T>C, het	rs1801265 (*9A)
	c.496A>G, het	rs2297595
	c.1129–5923C>G, het	rs75017182 (HapB3)
	c.1236G>A, het	rs56038477 (HapB3)
	c.2846A>T, het	rs67376798
HG00129	c.85T>C, het	rs1801265 (*9A)
	c.1129–5923C>G, het	rs75017182 (HapB3)
	c.1236G>A, het	rs56038477 (HapB3)
	c.2194G>A, het	rs1801160 (*6)
HG00185	c.1627A>G, het	rs1801159 (*5)
	c.1905+1G>A, het	rs3918290 (*2A)
HG00315	c.85T>C, het	rs1801265 (*9A)
	c.775A>G, het	rs45589337
	c.2194G>A, het	rs1801160 (*6)
HG00325	Exon 4 deletion, het	NA
HG00332	c.1679T>G, het	rs55886062 (*13)
HG00537	c.910T>C, het	rs183105782
	c.1627A>G, het	rs1801159 (*5)
	c.1896T>C, het	rs17376848
HG00538	c.910T>C, het	rs183105782
	c.1627A>G, het	rs1801159 (*5)
	c.1896T>C, het	rs17376848
HG00613	c.1314T>G, het	rs186169810
	c.1627A>G, het	rs1801159 (*5)
HG01253	c.85T>C, het	rs1801265 (*9A)
	c.775A>G, het	rs45589337
	c.1120C>T, het	rs201785202
	c.1896T>C, het	rs17376848
HG01350	c.1601G>A, het	rs1801158 (*4)
	c.1627A>G, het	rs1801159 (*5)
	c.1896T>C, het	rs17376848
HG01374	c.1120C>T, hom	rs201785202
	c.1896T>C, hom	rs17376848
HG01631	c.299_302del, het	rs72549309 (*7)
	c.1627A>G, het	rs1801159 (*5)
HG02645	c.85T>C, hom	rs1801265 (*9A)

Coriell ID	Consensus Genotype (NM_000110.4) [^]	rsID ^{\$} (Legacy Name)
	c.868A>G, het	rs146356975
	c.1218G>A, het	rs61622928
	c.1358C>G, het	rs144395748
HG02772	c.85T>C, het	rs1801265 (*9A)
	c.868A>G, het	rs146356975
	c.1218G>A, het	rs61622928
	c.1358C>G, het	rs144395748
	c.3067C>A, het	rs114096998
HG03115	c.61C>T, het	rs72549310
	c.85T>C, het	rs1801265 (*9A)
	c.1371C>T, het [#]	rs57918000
HG03645	c.1896T>C, het	rs17376848
	c.2279C>T, het	rs112766203
HG03649	c.85T>C, hom	rs1801265 (*9A)
	c.496A>G, hom	rs2297595
HG03716	c.85T>C, het	rs1801265 (*9A)
	c.2279C>T, het	rs112766203
HG03770	c.2657G>A, het	rs1801267 (*9B)
HG03787	c.85T>C, het	rs1801265 (*9A)
	c.2043_2058del, het	rs773499329
NA06991	c.1601G>A, het	rs1801158 (*4)
	c.2846A>T, het	rs67376798
NA12248	c.1627A>G, het	rs1801159 (*5)
	c.1679T>G, het	rs55886062 (*13)
NA18956	c.1774C>T, het	rs59086055
NA18966	c.1896T>C, het	rs17376848
	c.2678A>G, het	rs188052243
NA19093	c.85T>C, het	rs1801265 (*9A)
	c.1218G>A, hem [#]	rs61622928
	c.1627A>G, het	rs1801159 (*5)
	Exon 11 deletion, het	NA
NA19207	c.85T>C, het	rs1801265 (*9A)
	c.557A>G, het	rs115232898
NA19711	c.85T>C, hom	rs1801265 (*9A)
	c.1024G>A, het	rs183385770
	c.1627A>G, hom	rs1801159 (*5)
NA19900	c.85T>C, het	rs1801265 (*9A)
	c.1054C>G, het	rs190577302
	c.1371C>T, het	rs57918000

Coriell ID	Consensus Genotype (NM_000110.4) [^]	rsID ^{\$} (Legacy Name)
	c.1896T>C, het	rs17376848
NA19902	c.85T>C, het	rs1801265 (*9A)
	c.496A>G, het	rs2297595
	c.1054C>G, het	rs190577302
	c.1896T>C, het	rs17376848
NA20362	c.85T>C, het	rs1801265 (*9A)
	c.557A>G, het	rs115232898
	c.1129–5923C>G, het	rs75017182 (HapB3)
	c.1236G>A, het	rs56038477 (HapB3)
	c.2194G>A, het	rs1801160 (*6)
NA20786	c.1905C>G, het	rs3918289
NA20901	c.1905+1G>A, het	rs3918290 (*2A)

[^] RefSeq <https://www.ncbi.nlm.nih.gov/refseq/>, last accessed 5/17/2024

^{\$} dbSNP (<https://www.ncbi.nlm.nih.gov/snp>, last accessed 2/2/2024)

[#] variant not independently confirmed

NA, not applicable; hom, homozygous; het, heterozygous